

Are We Feeling the Heat?

General Disaster Exposure versus Hazard-Specific Experience as Predictors of Climate

Policy and Attribution Beliefs Across U.S. Counties



QSS 20 Final Project

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Abstract:

Does county-level exposure to federally declared natural disasters predict public support for climate policy and belief that global warming is affecting specific hazards, and do any such relationships persist once county partisanship is held constant? Using data on 3,144 United States counties, this paper tests whether county-level exposure to federally declared natural disasters between 2021 and 2025 predicts support for regulating carbon dioxide and the belief that global warming is affecting specific hazards. Opinion estimates are used from the Yale Climate Opinion Maps 9th edition. County-level 2024 presidential returns from the MIT Election Data Science

Lab are used to eventually control for partisanship. Two ideas are compared: a generic one that pairs total disaster counts with general policy support, and a hazard-matched one that pairs hurricane, flood, and fire exposure with the belief about that same hazard. The generic pairing yields no meaningful relationship, even once controlling for partisanship. The matched pairings yield positive relationships that shrink significantly but still exist once controlling for partisanship. Across all models, partisanship explains the large majority of the variation in county climate opinion, while disaster exposure adds less than one percentage point. Local experience appears to adjust specific hazard beliefs at a marginal level without altering policy positions that are clearly anchored to party beliefs.

Introduction:

This paper examines the relationship between natural disaster exposure, climate policy beliefs, and attributing climate change to natural disasters. I am attempting to understand if there is a correlation between personal lived experience and policy beliefs on the county level. Research clearly predicts that in the coming years and decades, as the effects of climate change exacerbate, natural disasters will increase both in frequency and severity (IPCC 2021). Following this logic, this paper is essentially attempting to understand if the IPCC finding can also illustrate that over the course of the coming years and decades, support for climate change reduction policy, which for the sake of this paper will be simplified to carbon dioxide emissions reduction policy, will increase.

People draw on personal and local experience as evidence in daily life. Humans naturally make use of the world around them to shape their perspective and understanding of issues. Past studies demonstrate that this is the case for climate change. Egan and Mullin's 2012 paper

demonstrated that personal experience can shift climate beliefs, anchoring experiential learning with the idea of climate change. However, the effects Egan and Mullin found were small and decayed very quickly over time (Egan and Mullin 2012). My research is moving past Egan and Mullin in a multitude of ways. First, they specifically examine increased average temperature, whereas I am examining the effect of natural disasters. Temperature increase is able to be measured uniformly across the county, whereas natural disasters are isolated incidents, and many of the counties I examined simply have not experienced one in the last five years, which was the scope of my data set.

Furthermore, I also attempted to specifically examine these relationships when partisanship is held constant. Egan and Mullin's study has traditional survey covariates, one of which was party identification and ideology, but they did not examine exactly how the relationship changes when partisanship is included versus excluded. Past studies have demonstrated that political identity, not information, tends to govern how Americans perceive climate risk. Kahan et al. 's 2012 paper found that higher science literacy and numeracy do not raise concern about climate risk and instead, they widen the gap between cultural and political groups. Kahan et al. illustrated how climate beliefs are anchored to partisanship and ambiguous environmental signals are assimilated to that anchor rather than weighed against it. Directly experiencing a natural disaster is one such signal: whether a flood or a wildfire is read as evidence of climate change depends on what the observer already believes (Kahan et al. 2012). I tested this expectation by directly comparing the relationship between natural disaster experience and opinions on climate policy with and without partisanship as an additional variable on the county level.

On top of testing the direct effect of controlling for partisanship, this paper also addresses other gaps in past research. Many papers, including Egan and Mullin's 2012 study, simply examine a variable to look at generic exposure and its effect on a generic outcome, like policy support. The second half of this statistical analysis examines direct attribution: how being exposed to specific disasters affects perception of whether that disaster was caused by climate change. Once again, I look at the effect both before and after partisanship is controlled to view its effect on the relationship.

This study examines all of these questions at the United States county level, using Yale Climate Opinion Maps (YCOM) for opinion and attribution variables, Federal Emergency Management Agency (FEMA) declared natural hazard disaster data from 2021 to 2025 for exposure quantification, and MIT Election Data Science Lab (MEDSL) 2024 county presidential returns as a partisanship control. We employ ordinary least squares and bivariate analysis, adding Democratic vote share and conducting an R-squared decomposition to isolate the effects of partisanship on these relationships.

The final research question was: does county-level exposure to federally declared natural disasters predict public support for climate policy and belief that global warming is affecting specific hazards, and do any such relationships persist once county partisanship is held constant? Before beginning analysis, I drew three hypotheses. One, generic disaster exposure would positively predict general climate policy support at the county level. Two, hazard-matched exposure predicts the matching attribution belief, and that this relationship will be stronger than generic pairing. Finally, the relationships discussed in the first and second hypotheses will decrease in size but remain once partisanship is controlled.

Analysis revealed that hypothesis one was incorrect; contrary to expectations, a slight inverse relationship emerged. Hypothesis two was supported: a positive relationship existed between experiencing a specific natural disaster and attributing climate change to that disaster, and this relationship was significantly stronger than the generic pairing. Hypothesis three received support. After controlling for partisanship, the generic pairing showed a null effect, and the direct attribution models exhibited significantly weakened relationships.

In the remainder of this paper, I will provide a comprehensive data and methodology overview followed by a more in-depth display and discussion of results and limitations. I will conclude with a discussion of these results and suggestions for future research. All graphs and tables from all regressions are located in the appendix of the paper.

Data:

This section provides a comprehensive overview of the data used and how the merged dataset was created. All datasets used county as their unit of analysis. The United States has 3,144 counties in the 50 states plus the District of Columbia. This was the N value for the first set of regressions. Once controlling for partisanship, the N value shrank to 3,122, as that was the number of counties I was able to cleanly merge with the MEDSL data. I made use of the 2025 estimates from the 9th edition of the YCOM, disaster exposure data from 2021 to 2025 from FEMA, and MEDSL 2024 presidential results. Of the massive YCOM dataset, I specifically examined four county-level outcomes: support for carbon dioxide regulation, as my proxy for larger climate policy support, and belief that climate change is directly affecting hurricanes, flooding, and wildfires. Four variables were examined from the FEMA dataset: total declared natural-hazard disasters, 2021 to 2025, plus hurricane, flood, and fire-specific counts. The one

variable used from the MEDSL dataset, as a proxy for political polarization, was the county's 2024 Democratic presidential vote share.

YCOM is not a collection of survey responses; instead, it is county-level modeled estimates for climate attitudes from the Yale Program on Climate Change Communication. The models are responses of multilevel regression and post-stratification, which combines national survey responses with census demographics to estimate opinions in places that are never directly polled (Howe et al. 2015). The YCOM data is a model output, not a local poll, which places a large asterisk on understanding the results of my paper. Estimates are percentage results from 0 to 100, with a reported margin of error between seven and nine percent.¹ Yale notes that the underlying model has changed over time, so cross-year comparisons are not purely attitudinal. This is why I am limited to looking at the 9th edition of the YCOM data, examining estimates from 2021-2025. I used the variable “regulate” as my policy option variable, and “attrhurricanes,” “attrflood,” and “attrwildfires” as my attribution variables. I chose these four specific variables I am examining to answer both my general policy support research question along with the specific attribution question I examine.

The source of the natural disaster data was from the OpenFEMA Disaster Declarations Summaries v2. I used the variable "disaster_count" for the first regression and “disaster_count_Hurricane,” “disaster_count_Flood,” and “disaster_count_Fire” for the specific values for each type of disaster. These variables were then matched with the aforementioned attribution type variables from the YCOM data set.

¹ It is worth noting that while editing this paper I discovered YCOM claims to control for partisanship in their methodology (Mildenberger et al. 2020). I have added a commentary to this in the limitation section of my discussion.

Finally, the source of the partisanship control was the MEDSL official precinct-level 2024 release, aggregated to the county level and filtered to the United States Presidential contest. I use the “per_dem” variable to examine partisanship. Note the “per_gop” and “per_dem” variables are correlated at $r = -0.999$, so only one was needed. In total, 3,122 of the 3,144 counties matched the previous two data sets. The main gap existed in Alaska, as it does not report at the county level.

The data were merged across the three tables with an inner join on county FIPS codes. The final sample size was 3,144 counties for the bivariate models, and 3,122 for the models in which partisanship was controlled.

Methods:

This experiment is split into two models: the bivariate model, model one, and then the multivariate analysis in which partisanship is controlled, model two. Both models are ordinary least squares (OLS) on the county level. Each model is a single hazard outcome pair estimated separately, meaning there is no pooling across models and outcomes.

Model One is bivariate and tests hypotheses one and two. The model uses the equation $Y_c = \beta_0 + \beta_1 \cdot \text{Disaster}_c + \varepsilon_c$. Y_c is the outcome for the given county c . Disaster_c is the predictor for county c , so either the “disaster_count” or the specific type of hurricane, flood, or fire. β_0 is the intercept; the model predicts the outcome for a county with zero declared disasters of that type. β_1 is the slope; the estimated change in outcome, in percentage points of county opinion, associated with one additional declared disaster. ε_c is the error term. Hypothesis one looks at how the general disaster count affects belief about general carbon dioxide policy,

whereas hypothesis two matches the specific disaster type with specific attribution of that disaster being caused by climate change.

Model Two is multivariate and tests hypothesis three. The model used the question $Y_c = \beta_0 + \beta_1 \cdot \text{Disaster}_c + \beta_2 \cdot \text{DemVoteShare}_c + \varepsilon_c$. The equation terms are the same as model one, except β_1 is now the estimated change in outcome per additional declared disaster while holding partisanship constant. Additionally, DemVoteShare_c is the “per_dem” variable illustrating the Democratic candidate’s share of the county c’s 2024 presidential votes. β_2 is the estimated change in outcome per one-unit increase in Democratic vote share, holding disaster exposure constant. Because the variable “per_dem” runs from 0 to 1, β_2 is the predicted swing in county opinion from an all Republican to an all-Democratic county.

To analyze Model Two, I made use of an R-squared decomposition. For each Model Two, I ran an additional regression on only “per_dem,” seeing the effect of “per_dem” on Y alone, and recorded its R^2 . I then found ΔR^2 by taking the R^2 of the whole model and subtracting the R^2 of just partisanship. The ΔR^2 is thus the variance the disaster variable explains on top of the partisanship. To visualize each controlled relationship, I residualize both the disaster count and the outcome of “per_dem” and plot the residuals with an OLS fit.

To reproduce my results, a person would have to download my scripts and data from my GitHub repository and run them in a Python environment that imports “pandas,” “statsmodels,” and “matplotlib.”

Results:

Throughout this section of the paper, I will go over the results of my analysis, diving into both Model One and Model Two for both the general and hazard-specific cases. Full tabulated results are visible in the appendix.

Table One illustrates the bivariate results. The generic pair, between disaster exposure and climate policy belief, yields results that were contrary to hypothesis one. A slight statistically significant, at $p < 0.005$, negative relationship was found with a slope of -0.110 and an R^2 of 0.003. The statistical significance is only due to the large sample size of 3,144; in actuality, the effect is negligible. Hypothesis one was not supported. All three matched pairs demonstrated positive correlations that were statistically significant at $p < 0.001$. First, in the hurricane case, there was a slope of 1.097 and R^2 of 0.024. Second, the flood case had a slope of 1.477 and an R^2 of 0.009. Finally, the fire case had a slope of 2.246 and an R^2 of 0.016. These results demonstrate that matching the hazard to the belief turns null or negative into three consistent positive associations. Effect sizes are small, with R^2 values of one to two percent, but are all significant. Therefore, hypothesis two is supported.

Table Two depicts the multivariate with R-squared decomposition results. For the generic pairing, once partisanship is controlled, the slope shrinks from -0.110 towards zero, with a new ΔR^2 of essentially 0.000. For the matched pairs, all relationships survive but significantly shrink, having ΔR^2 values of: 0.0026 for the hurricane case, 0.0002 for the flood case, and 0.0051 for the fire case. It is important to note that the flood case barely survived, and a ΔR^2 value of 0.0002 is essentially no relationship. The decomposition illustrates that partisanship alone explains roughly 77 to 89 percent of the cross-county variance; adding disaster exposure raised the R^2 by 0.0002 to 0.0051. Partisanship clearly dominates the relationship, and disaster exposure only adds a

small statistically robust increment. All three matched pairs relationships decreased in size once partisanship was added to the question and therefore, hypothesis three is supported.

Taken together, the results reject hypothesis one and support hypotheses two and three; generic disaster exposure is essentially unrelated to policy support, while hazard-matched exposure predicts the matching attribution belief, and this relationship attenuates but largely holds under the partisanship control. Even so, partisanship dominates every model. This is merely a summary of results. For all results, please look for the appendix which includes all regression tables and figures depicting all of the regressions.

Discussion:

The results of my analysis demonstrate that, net of partisanship, disaster exposure has no relationship with general climate policy support and a small, hazard-specific relationship with attribution belief. Overall, throughout this paper I have shown that, on the county level, once controlling for partisanship, whether I live in an area constantly affected by hurricanes, floods, and wildfires will not change my support for climate change policy and only slightly alter my belief that climate change caused those hurricanes, floods, and wildfires. Throughout all models, partisanship is the dominant predictor.

These results lie between the two accounts presented in the introduction of this paper. The experiential learning perspective, discussed in Egan and Mullin's 2012 paper, predicts that direct exposure to hazards updates climate belief. And it receives partial support from my findings. Counties that have been hit by hurricanes or wildfires are modestly more likely to say that global warming is affecting those specific hazards, even after partisanship is held constant. But the results are very small and narrow. Exposure moved the specific factual belief closest to

the experience, not the broader question of whether to support climate policy, which shows no independent relationship with disaster history. The motivated reasoning argument that Kahan et al's 2012 paper discussed, on which partisanship structures how ambiguous evidence is interpreted, better describes the overall pattern in my results. Partisanship explains the large majority of the variation in county climate opinion, and disaster exposure operates only in the small residuals. Taken together, local experience appears to nudge concrete hazard-attribution beliefs at the margin without reshaping the policy positions that are firmly anchored to the party.

The contrast between the generic pairing and the hazard-matched models is itself an interesting finding. Pairing a county's total disaster count with general policy support, an intuitive but coarse design that is common in prior work, returns nothing: a near-zero, wrong-signed coefficient that vanishes once partisanship is controlled. Pairing each hazard type with the belief about that same hazard, by contrast, recovers positive, significant associations that the generic specification does not detect. This suggests that some of the inconsistent results in the disaster-and-opinion literature may be right that exposure does not move policy support, while still missing the hazard-specific beliefs that do respond to local experience. The distinction matters because it locates the effect: where disaster exposure registers at all, it is on concrete attribution beliefs rather than on broader policy preferences, and only when the hazard is matched to the belief being measured. This is a narrow claim, and not one about climate policy, but it points to a local, place-based effect that is invisible when only the large policy question is asked.

It is important to understand that all relationships that survive the partisanship control are very modest. Across all four outcomes, county partisanship accounts for 77 to 89 percent of the variation in climate opinion, while adding disaster exposure only raises the explained variance by

a max of 0.51 percentage points. In the strongest case, wildfires, disaster exposure explained less than one-hundredth as much variation as partisanship does. The substantive picture is therefore one in which disaster exposure is a small nudge to hazard-specific beliefs while partisanship determines county beliefs overall. The effect is still notable, since it survives a demanding control in a consistent pattern, but its weight is significantly limited.

Limitations and Future Research:

There are several limitations to these results. First, the analysis is on the county level and not on the individual level. It is cross-sectional and observational, so the associations are not causal estimates; counties were not randomly assigned their disaster history, and, as the partisanship control demonstrates, citizens may sort across places in a way correlated with climate views. Second, the outcome variables are based estimated using YCOM, with a reported margin of error of roughly seven to nine percentage points, so small between-county differences should not be overstated. Third, disaster exposure is measured by federal declarations, which capture what received a declaration rather than raw hazard incidents and undercount slower or smaller events that were never declared or exist between a clear disaster type. Fourth, my model only included one control; partisanship is the most important confounder, but income, education, urbanity, region, and baseline physical risk are plausibly related to both exposure and opinion, and therefore, their omission leaves room for bias. Finally late in the research process I learned that YCOM attempts to control for partisanship in its internal model (Mildenberger et al. 2017). While I initially thought this discredited my analysis, after a deeper dive I concluded that, assuming YCOM does control for partisanship, my findings show that the control is not

complete. I now believe my results only establish a floor for the magnitude of partisanship in the relationship.

My results leave many questions unanswered and can serve as a guide for future research. For example, if the YCOM data set is expanded to include other types of disasters, like droughts or tropical storms, you could understand if this correlation exists across a wider range of disasters. A second possible area of study could be a randomized field study in which respondents were asked policy and attribution questions specifically, and answering demographic data about natural hazards could provide more individual-level data and the possibility for a causal claim. Finally, a difference-in-differences event study in an area prone to a specific disaster type, like wildfires in Colorado, could show the longevity of any effects.

Conclusion:

This paper asked: does county-level exposure to federally declared natural disasters predict public support for climate policy and belief that global warming is affecting specific hazards, and do any such relationships persist once county partisanship is held constant? The answer divides along those two outcomes. Once partisanship is controlled, exposure has no detectable relationship with general policy support, but it retains a small relationship with hazard-specific attribution belief. That second effect is real; it survives the demanding partisanship control, and it appears only when the hazard is matched to the belief being asked about. However, it is minor next to partisanship, which explains most of the variation in county climate opinion. The broader takeaway from my analysis is that how a county votes predicts its climate view far better than recent drastic weather. Direct experience appears to augment specific

factual beliefs at a marginal level without actually altering policy positions. Climate policy remains anchored to party view.

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